

Frequency-domain features for motor condition monitoring

From a time waveform to machine-learning inputs with FFT and Goertzel

A lecture deck with a runnable Python notebook

Illustrative signals throughout

Why frequency features help

- Motor faults often modulate a rotating or electrical carrier.
- The time trace mixes useful periodic structure with noise and operating changes.
- A spectrum concentrates repeating energy at frequencies that can become model features.

The learning task

**waveform → features →
condition label**

Examples: healthy, imbalance, bearing damage

A motor signal contains several frequency families

Mechanical speed, electrical supply, harmonics, sidebands and broadband noise can coexist

1× speed

shaft rotation

2× speed

misalignment

line frequency

current signal

harmonics

nonlinearity

bearing band

resonance

Feature engineering starts with physics: choose frequency regions that plausibly change with the fault and remain stable enough to measure.

- Vibration commonly exposes mechanical and bearing resonances.
- Motor current can reveal supply, rotor and load-related modulation.
- Tachometer speed or mains frequency helps normalize operating changes.

Sampling sets the frequency resolution

Key quantities

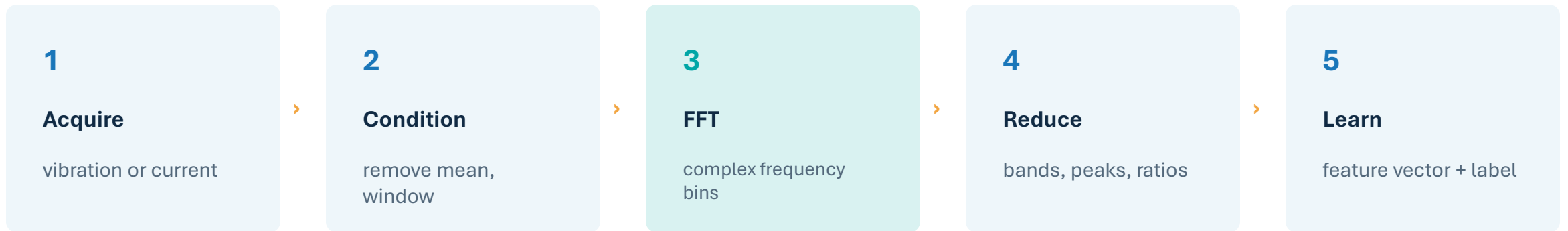
Nyquist frequency: $f_n = f_s / 2$

Bin spacing: $\Delta f = f_s / N = 1 / T$

Example: 12 kHz for 1 s gives $\Delta f = 1$ Hz

- Sample fast enough to capture the highest feature band.
- Use a longer record when close frequencies must separate.
- Apply an anti-alias filter before digitizing.
- Keep sampling settings consistent across training and deployment.

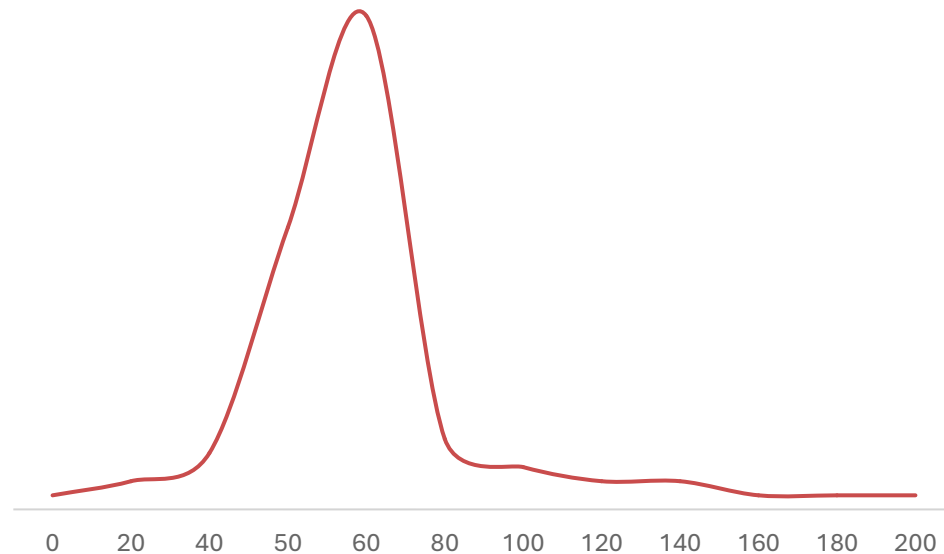
FFT workflow for one measurement window



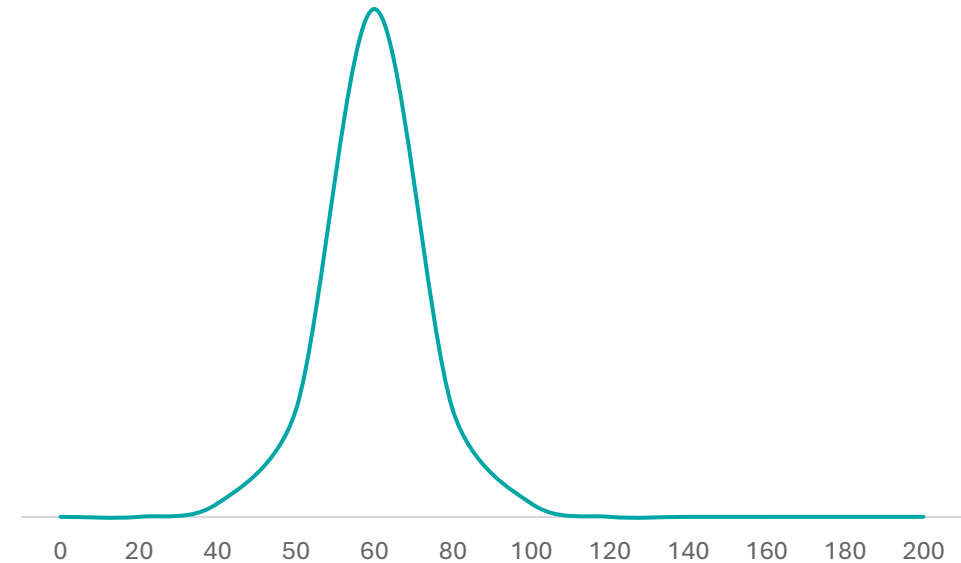
Use fixed-length windows and the same preprocessing for every record. Preserve the scaling and feature definitions in your model pipeline.

Windowing reduces spectral leakage

A finite record rarely begins and ends at a waveform zero crossing



Hard edges spread energy across nearby bins



A Hann window lowers leakage at the cost of peak width

Feature recipes from an FFT magnitude spectrum

Peak amplitude

max amplitude near an expected frequency

Band power

sum of squared magnitudes in a frequency interval

Harmonic ratio

2× or 3× energy relative to the fundamental

Sideband energy

energy near a carrier $\pm$ modulation frequency

Spectral shape

centroid, bandwidth, kurtosis or entropy

Prefer a small set tied to a diagnostic question over thousands of unexamined bins.

Goertzel targets a small set of known frequencies

When it fits

- You need a few tones or harmonics.
- The targets are known in advance.
- You want low memory or embedded execution.

What it computes

A recurrence evaluates the DFT at selected frequency bins instead of building the full spectrum.

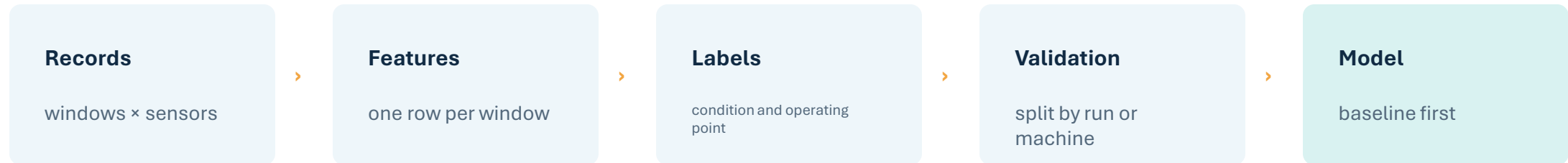
It is a feature extractor, not a replacement for exploration.

FFT and Goertzel serve different measurement questions

Question	FFT	Goertzel
Explore unknown patterns	Strong fit	Poor fit
Feature count	Many bins or bands	A few targets
Compute cost	One full transform	Scales with targets
Best use	Discovery and rich features	Embedded monitoring

Common approach: use FFT during development, then deploy Goertzel for proven frequency features when resources are limited.

From feature matrix to a reliable model



- Split by machine, operating run or time block to avoid leaking nearly identical windows into train and test.
- Normalize features using training data only.
- Inspect feature importance and errors at each operating condition.

Takeaway

Use the FFT to discover and summarize frequency structure. Use Goertzel when a small, stable group of frequencies has already earned its place in the feature set.

Open the companion notebook and change the simulated fault amplitude.

Then compare peak, band-power and Goertzel features before building a classifier.

Questions for the room: which frequency moves when speed changes?